

The Problem

The company's **profit** is given by this equation:

$$P(x_1, x_2) = x_1^3 + x_2^3 + 2x_1^2 + 4x_2^2 + 6,$$

where:

- x_1 : salary of employees.
- x_2 : general expenses.

We want to:

1. **Find the stationary points** (where the profit stops increasing or decreasing).
 2. **Check the nature** of these points (do they give maximum profit, minimum profit, or something else?).
 3. **Find the values of x_1 and x_2** where the profit is maximized.
-

Step 1: What Are Stationary Points?

Stationary points are where the **profit stops changing**. This happens when:

$$\frac{\partial P}{\partial x_1} = 0 \quad \text{and} \quad \frac{\partial P}{\partial x_2} = 0.$$

These are the **partial derivatives** of P , which tell us how P changes with respect to x_1 and x_2 .

Step 2: Find the Partial Derivatives

Let's calculate the partial derivatives:

1. **Partial derivative with respect to x_1 :** Take the derivative of $P(x_1, x_2)$ while treating x_2 as constant:

$$\frac{\partial P}{\partial x_1} = 3x_1^2 + 4x_1.$$

2. **Partial derivative with respect to x_2 :** Take the derivative of $P(x_1, x_2)$ while treating x_1 as constant:

$$\frac{\partial P}{\partial x_2} = 3x_2^2 + 8x_2.$$

Step 3: Solve for Stationary Points

To find stationary points, set both partial derivatives to **zero**.

Solve for x_1 :

$$3x_1^2 + 4x_1 = 0.$$

Factorize:

$$x_1(3x_1 + 4) = 0.$$

Solutions:

$$x_1 = 0 \quad \text{or} \quad x_1 = -\frac{4}{3}.$$

Solve for x_2 :

$$3x_2^2 + 8x_2 = 0.$$

Factorize:

$$x_2(3x_2 + 8) = 0.$$

Solutions:

$$x_2 = 0 \quad \text{or} \quad x_2 = -\frac{8}{3}.$$

Stationary Points:

The stationary points are combinations of x_1 and x_2 :

$$(x_1, x_2) = (0, 0), (0, -\frac{8}{3}), (-\frac{4}{3}, 0), (-\frac{4}{3}, -\frac{8}{3}).$$

Step 4: Nature of Stationary Points (Hessian Matrix)

The **Hessian matrix** helps us determine whether a stationary point is a maximum, minimum, or saddle point. The Hessian is a table of second derivatives of $P(x_1, x_2)$:

$$H(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 P}{\partial x_1^2} & \frac{\partial^2 P}{\partial x_1 \partial x_2} \\ \frac{\partial^2 P}{\partial x_2 \partial x_1} & \frac{\partial^2 P}{\partial x_2^2} \end{bmatrix}.$$

Second Derivatives:

1. $\frac{\partial^2 P}{\partial x_1^2} = 6x_1 + 4,$
2. $\frac{\partial^2 P}{\partial x_2^2} = 6x_2 + 8,$

3. $\frac{\partial^2 P}{\partial x_1 \partial x_2} = 0$ (no cross-terms).

Hessian Matrix:

$$H(x_1, x_2) = \begin{bmatrix} 6x_1 + 4 & 0 \\ 0 & 6x_2 + 8 \end{bmatrix}.$$

Step 5: Determine the Nature of Each Stationary Point

The eigenvalues (diagonal elements) of $H(x_1, x_2)$ are:

$$\lambda_1 = 6x_1 + 4, \quad \lambda_2 = 6x_2 + 8.$$

We check whether both eigenvalues are positive (positive definite), negative (negative definite), or mixed (indefinite):

1. At $(0, 0)$:

- $\lambda_1 = 6(0) + 4 = 4$ (positive),
- $\lambda_2 = 6(0) + 8 = 8$ (positive).

Both are positive, so this is a **local minimum**.

2. At $(0, -\frac{8}{3})$:

- $\lambda_1 = 6(0) + 4 = 4$ (positive),
- $\lambda_2 = 6(-\frac{8}{3}) + 8 = -8 + 8 = 0$.

One eigenvalue is zero, so this is a **saddle point** (neither max nor min).

3. At $(-\frac{4}{3}, 0)$:

- $\lambda_1 = 6(-\frac{4}{3}) + 4 = -8 + 4 = -4$ (negative),

- $\lambda_2 = 6(0) + 8 = 8$ (positive).

Mixed signs indicate a **saddle point**.

4. At $(-\frac{4}{3}, -\frac{8}{3})$:

- $\lambda_1 = 6(-\frac{4}{3}) + 4 = -8 + 4 = -4$ (negative),
- $\lambda_2 = 6(-\frac{8}{3}) + 8 = -16 + 8 = -8$ (negative).

Both are negative, so this is a **local maximum**.

Step 6: Maximize the Profit

The **maximum profit** occurs at $(x_1, x_2) = (-\frac{4}{3}, -\frac{8}{3})$.

Substitute into $P(x_1, x_2)$:

$$P(-\frac{4}{3}, -\frac{8}{3}) = (-\frac{4}{3})^3 + (-\frac{8}{3})^3 + 2(-\frac{4}{3})^2 + 4(-\frac{8}{3})^2 + 6.$$

1. Calculate each term:

- $(-\frac{4}{3})^3 = -\frac{64}{27},$
- $(-\frac{8}{3})^3 = -\frac{512}{27},$
- $2(-\frac{4}{3})^2 = \frac{32}{9} = \frac{96}{27},$
- $4(-\frac{8}{3})^2 = \frac{256}{9} = \frac{768}{27},$
- Constant term = 6.

2. Add them:

$$P = -\frac{64}{27} - \frac{512}{27} + \frac{96}{27} + \frac{768}{27} + 6.$$

3. Combine:

$$P = \frac{288}{27} + 6 = \frac{450}{27} = 16.67.$$

Final Answer

The maximum profit is:

$$\boxed{16.67},$$

which occurs at:

$$(x_1, x_2) = \left(-\frac{4}{3}, -\frac{8}{3}\right).$$